Predictor corrector method

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Milne's Predictor Corrector Method

Adam's k step explicit method →

$$y_{n+1} = y_{n-l} + \sum_{m=0}^{k-1} \beta_m^{(l)} \nabla^m f_n$$

where $\beta_m^{(l)} = \int_{-l}^{1} (-1)^m \binom{-s}{m} ds = \int_{-l}^{1} \frac{s(s+1)\cdots(s+m-1)}{m!} ds$

Taking $l = 3$, $k = 4$, we get

$$y_{n+1} = y_{n-3} + \sum_{m=0}^{3} \beta_m^{(3)} \nabla^m f_n$$

$$\beta_m^{(3)} = \int_{-3}^{1} \frac{s(s+1)\cdots(s+m-1)}{m!} ds \qquad m = 0, 1, 2, 3$$

$$\beta_0^{(3)} = \int_{-3}^{1} ds = 4$$

$$\beta_1^{(3)} = \int_{-3}^{1} s\,ds = \frac{1-9}{2} = -4$$

$$\beta_2^{(3)} = \int_{-3}^{1} \frac{s(s+1)}{2} ds = \int_{-3}^{1} \frac{s^2+s}{2} ds = \left(\frac{1+27}{6} + \frac{1-9}{4}\right)$$

$$= \frac{\cancel{28}^{14}}{\cancel{6}_{3}} - 2$$

$$= \frac{8}{3}$$

$$\beta_3^{(3)} = \int_{-3}^{1} \frac{s(s+1)(s+2)}{6} ds = \int_{-3}^{1} \frac{s^3+3s^2+2s}{6} ds$$

$$= \frac{1-81}{24} + \frac{1+27}{6} + \frac{1-9}{6}$$

$$= -\frac{\cancel{80}^{10}}{\cancel{24}_{3}} + \frac{\cancel{28}^{14}}{\cancel{6}_{3}} - \frac{\cancel{8}^{4}}{\cancel{6}_{3}}$$

$$= 0$$

m	0	1	2	3
$\beta_m^{(3)}$	4	-4	8/3	0

$$y_{n+1} = y_{n-3} + h\left[4f_n - 4\nabla f_n + \frac{8}{3}\nabla^2 f_n\right]$$

$$= y_{n-3} + h\left[\cancel{4f_n} - 4(\cancel{f_n} - f_{n-1}) + \frac{8}{3}(f_n - 2f_{n-1} + f_{n-2})\right]$$

$$= y_{n-3} + h\left[\frac{8}{3}f_n - \frac{4}{3}f_{n-1} + \frac{8}{3}f_{n-2}\right]$$

$$\boxed{y_{n+1} = y_{n-3} + \frac{4}{3}h\left[2f_n - f_{n-1} + 2f_{n-2}\right]}$$

This is called Milne's Predictor formula. It is of order 4.

Milne's corrector formula → Adan's k step implicit formula is given by

$$y_{n+1} = y_{n-l} + \sum_{m=0}^{k} \gamma_m^{(l)} \nabla^m f_{n+1}$$

$$\text{where} \quad \gamma_m^{(l)} = \int_{-l}^{1} (-1)^m \binom{1-s}{m} ds = \int_{-l}^{1} \frac{(s-1)s\ldots(s+m-2)}{m!} ds$$

Take $l = 1$, $k = 3$

$$y_{n+1} = y_{n-1} + h\sum_{m=0}^{3} \gamma_m^{(1)} \nabla^m f_{n+1}, \quad \text{where}$$

$$\gamma_m^{(1)} = \int_{-1}^{1} \frac{(s-1)s\ldots(s+m-2)}{m!} ds \qquad m = 0, 1, 2, 3$$

$$\gamma_0^{(1)} = \int_{-1}^{1} ds = 2$$

$$\gamma_1^{(1)} = \int_{-1}^{1} (s-1)\, ds = \left(\frac{s^2}{2} - s\right)\Big|_{-1}^{1} = -2$$

$$\gamma_2^{(1)} = \int_{-1}^{1} \frac{(s^2 - s)}{2} ds = \frac{1}{3}$$

$$\gamma_3^{(1)} = 0$$

m	0	1	2	3
$\gamma_m^{(1)}$	2	-2	1/3	0

$$\boxed{y_{n+1} = y_{n-1} + \frac{h}{3}\left[f_{n+1} + 4f_n + f_{n-1}\right]}$$

This is called Milne's corrector formula. It is of order 4.

The Milne's P-C pair (Predictor - Corrector pair) is defined as

$$P : y_{n+1} = y_{n-3} + \frac{4}{3} h \left[2f_n - f_{n-1} + 2f_{n-2} \right]$$

$$C : y_{n+1} = y_{n-1} + \frac{h}{3} \left[f_{n+1} + 4f_n + f_{n-1} \right]$$

(looks like Simpson's one-third rule)

Predictor corrector schemes →

① PECE scheme - This is explained as follows →

i) P - Find an approx. for y_{n+1} using the predictor formula and denote it by $y_{n+1}^{(P)}$.

ii) E - Evaluate $f(x_{n+1}, y_{n+1}^{(P)}) = f_{n+1}^{(P)}$

iii) C - Find the new approx. for y_{n+1} by using $f_{n+1} = f(x_{n+1}, y_{n+1}^{(P)})$ on RHS of corrector formula.
Denote it by $y_{n+1}^{(C)}$.

iv) Evaluate $f_{n+1}^{(C)} = f(x_{n+1}, y_{n+1}^{(C)})$

We then go to the next step / iteration / nodal point.

Solution at x_{n+2} ⇒

$$P : y_{n+2}^{(P)} = y_{n-2} + \frac{4}{3} h \left[2f_{n+1}^{(C)} - f_n + 2f_{n-1} \right]$$

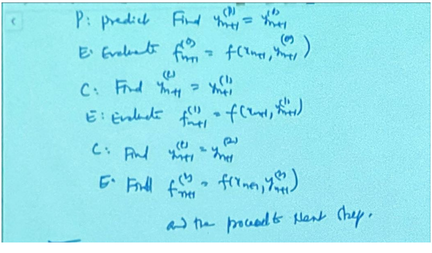
$$E : f_{n+2}^{(P)} = f(x_{n+2}, y_{n+2}^{(P)})$$

$$C : y_{n+2}^{(C)} = y_{n-2} + \frac{h}{3} \left[f_{n+2}^{(P)} + 4f_{n+1}^{(C)} + f_n \right]$$

$$E : f_{n+2}^{(C)} = f(x_{n+2}, y_{n+2}^{(C)})$$

Then go to the next step.

② $P(EC)^2E$ Scheme PECECE



P: predict Find $y_{n+1}^{(P)} = y_{n+1}^{(0)}$

E: Evaluate $f_{n+1}^{(0)} = f(x_{n+1}, y_{n+1}^{(0)})$

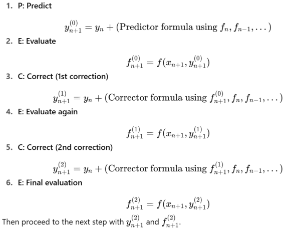
C: Find $y_{n+1}^{(C)} = y_{n+1}^{(1)}$

E: Evaluate $f_{n+1}^{(1)} = f(x_{n+1}, y_{n+1}^{(1)})$

C: Find $y_{n+1}^{(C)} = y_{n+1}^{(2)}$

E: Find $f_{n+1}^{(2)} = f(x_{n+1}, y_{n+1}^{(2)})$

and then proceed to next step.



1. **P: Predict**

$$y_{n+1}^{(0)} = y_n + (\text{Predictor formula using } f_n, f_{n-1}, \dots)$$

2. **E: Evaluate**

$$f_{n+1}^{(0)} = f(x_{n+1}, y_{n+1}^{(0)})$$

3. **C: Correct (1st correction)**

$$y_{n+1}^{(1)} = y_n + (\text{Corrector formula using } f_{n+1}^{(0)}, f_n, f_{n-1}, \dots)$$

4. **E: Evaluate again**

$$f_{n+1}^{(1)} = f(x_{n+1}, y_{n+1}^{(1)})$$

5. **C: Correct (2nd correction)**

$$y_{n+1}^{(2)} = y_n + (\text{Corrector formula using } f_{n+1}^{(1)}, f_n, f_{n-1}, \dots)$$

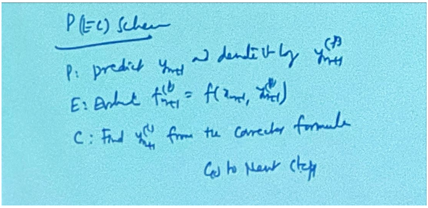
6. **E: Final evaluation**

$$f_{n+1}^{(2)} = f(x_{n+1}, y_{n+1}^{(2)})$$

Then proceed to the next step with $y_{n+1}^{(2)}$ and $f_{n+1}^{(2)}$.

③ PEC scheme

In this, for the next iteration, the evaluated value of fn+1 using yn+1(P) is used, instead of the one using yn+1(C) like in PECE.



P(EC) Scheme

P: predict y_{n+1} and denote it by $y_{n+1}^{(P)}$

E: Evaluate $f_{n+1}^{(P)} = f(x_{n+1}, y_{n+1}^{(P)})$

C: Find $y_{n+1}^{(C)}$ from the Corrector formula

Go to Next step

We also have PECEC = $P(EC)^2$ scheme.

Simplest predictor - corrector method

P : $y_{n+1} = y_n + hf(x_n, y_n)$ 1st order

C : $y_{n+1} = y_n + \frac{h}{2}[f_{n+1} + f]$ 2nd order

PECE →

P : $y_1^{(P)} = y_0 + hf(x_0, y_0)$

E : $f_1^{(P)} = f(x_1, y_1^{(P)})$

C : $y_1^{(C)} = y_0 + \frac{h}{2}[f_1^{(P)} + f_0]$

E : $f_1^{(C)} = f(x_1, y_1^{(C)})$

Proceed to next step.

Problem Use Milne's predictor - corrector method to obtain the solution of $y' = f(x, y)$ at $x = 1.2$ and $x = 1.5$ given that $y(0) = 0$, $y(0.3) = 0.04986$, $y(0.6) = 0.2221$, $y(0.9) = 0.5596$ using PECE scheme.
(If nothing is mentioned, PECE is default scheme).

* If initial values not given, use single step method to find them.

Solution P : $y_{n+1} = y_{n-3} + \frac{4}{3}h[2f_n - f_{n-1} + 2f_{n-2}]$

C : $y_{n+1} = y_{n-1} + \frac{1}{3}h[f_{n+1} + 4f_n + f_{n-1}]$

n = 3 P : $y_4 = y_0 + \frac{4h}{3}[2f_3 - f_2 + 2f_1]$

C : $y_4 = y_2 + \frac{h}{3}[f_4 + 4f_3 + f_2]$

$f(x, y) = x + y$ [given]

$f_1 = f(x_1, y_1) = x_1 + y_1 = 0.3 + 0.0499 = 0.3499$

$f_2 = 0.8221$

$f_3 = 1.4596$

$$P: \; y_4^{(P)} = 0 + \frac{4}{3} \times 0.3 \left(2 \times 1.4596 - 0.8221 + 2 \times 0.3499\right)$$

$$= 1.1187$$

$$E: \; f_4^{(P)} = f\left(x_4, y_4^{(P)}\right) = 1.2 + 1.1187$$

$$= 2.3187$$

$$C: \; y_4^{(c)} = 0.2221 + \frac{0.3}{3}\left[2.3187 + 4 \times 1.4596 + 0.8221\right]$$

$$= 1.1200$$

$$E: \; f_4^{(c)} = f\left(x_4, y_4^{(c)}\right) = 1.2 + 1.1200$$

$$= 2.3200$$

$$\boxed{y(1.2) \sim y_4^{(c)} = 2.3200}$$

Now, for $n = 4$ →

$$P: \; y_5^{(P} = y_1 + \frac{4h}{3}\left[2f_4^{(c)} - f_3 + 2f_2\right]$$

$$C: \; y_5^{(c)} = y_3 + \frac{h}{3}\left[f_5^{(P)} + 4f_4^{(c)} + f_3\right]$$

$f_2 = 0.8221$, $f_3 = 1.4596$, $f_4^{(c)} = f_4 = 2.3200$

$$P: \; y_5^{(P)} = 0.04987 + \frac{4 \times 0.3}{3}\left[2 \times 2.3200 - 1.4596 + 2 \times 0.8221\right]$$

$$= 1.9798$$

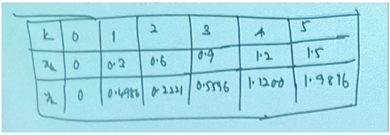
$$E: \; f_5^{(P)} = f\left(x_5, y_5^{(P)}\right) = 1.5 + 1.9798$$

$$= 3.4798$$

$$C: \; y_5^{(c)} = 0.5596 + \frac{0.3}{3}\left[3.4798 + 2.32 \times 2 + 1.4596\right]$$

$$= 1.9816$$

$$E: \; f_5^{(c)} = 1.5 + 1.9816 = 3.4816$$



k	0	1	2	3	4	5
x_k	0	0.3	0.6	0.9	1.2	1.5
y_k	0	0.4986	0.2221	0.5596	1.1200	1.9876

$y' = f(x, y)$ $\qquad\qquad y(x_0) = y_0$

$$\int_{x_{n-1}}^{x_{n+1}} y'\,dx = \int_{x_{n-1}}^{x_{n+1}} \underline{f(x, y(x))\,dx}$$

$\hookrightarrow$ Approximate by Simpson's $\frac{1}{3}$ rule

$$y(x_{n+1}) - y(x_{n-1}) \approx \frac{h}{3}\left[f(x_{n-1}, y_{n-1}) + 4f(x_n, y_n) + f(x_{n+1}, y_{n+1})\right]$$

$$\Rightarrow y_{n+1} = y_{n-1} + \frac{h}{3}\left[f_{n-1} + 4f_n + f_{n+1}\right]$$

This is Milne's corrector method (a 2 step method).

General linear k step method

$$\boxed{\sum_{j=0}^{k} \alpha_j y_{n+j} = h \sum_{j=0}^{k} \beta_j f(x_{n+j}, y_{n+j})} \qquad - ①$$

$\alpha_0, \alpha_1, \ldots \alpha_k$ and $\beta_0, \beta_1, \ldots \beta_k$ are real constants.

such that $\alpha_0^2 + \beta_0^2 \neq 0 \iff |\alpha_0| + |\beta_0| = 0$ [To ensure it is k step]

[$\beta_k = 0 \rightarrow$ explicit method, $\beta_k \neq 0 \rightarrow$ Implicit method] $\quad \alpha_k \neq 0$

Why linear? Uses only linear combinations of $\{y_n\}, \{f(x_n, y_n)\}$

Eg. 4 step Adams Bashforth method

$$y_{n+4} = y_{n+3} + \frac{1}{24} h\left[55 f_{n+3} - 59 f_{n+2} + 37 f_{n+1} - 9 f_n\right]$$

is an example of an explicit 4 step general linear method.

4 step Adam Moulton method

$$y_{n+4} = y_{n+3} + \frac{1}{24}\left[9 f_{n+4} + 19 f_{n+3} - 5 f_{n+2} - 9 f_{n+1}\right]$$

is an eg. of implicit 4 step method.

Zero stability

A linear k step method is said to be zero stable if ∃ a const. k s.t. for any 2 seq. $\{y_n\}$, $\{\tilde{y}_n\}$ which have been generated by the same formula but diff. initial data as $y_0, y_1, \dots y_{n-1}$ and $\tilde{y}_0, \tilde{y}_1 \dots \tilde{y}_{n-1}$ respectively, we have

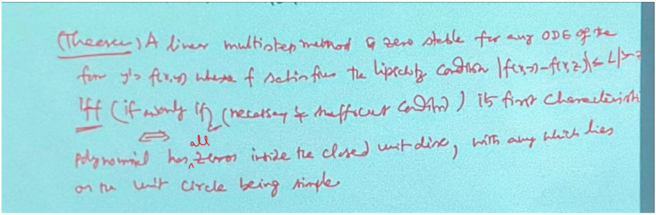
$$\boxed{|y_n - \tilde{y}_n| \leq M \max\{|y_0 - \tilde{y}_0|, |y_1 - \tilde{y}_1| \dots, |y_{k-1} - \tilde{y}_{k-1}|\}}$$

A general k step method in ①, consider its 1st & 2nd characteristic polynomial respectively.

$$\rho(z) = \sum_{j=0}^{k} \alpha_j z^j$$

$$\sigma(z) = \sum_{j=0}^{k} \beta_j z^j \qquad \text{where } \alpha_k \neq 0, \quad \alpha_0^2 + \beta_0^2 \neq 0$$

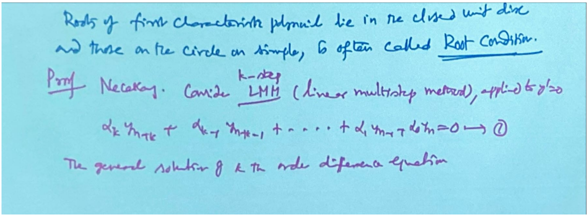
A linear multistep method is zero stable for any ODE of the form $y' = f(x, y)$ where f satisfies the Lipschitz condition $|f(x,y) - f(x,z)| \leq L|y - z|$ iff its 1st characteristic polynomial



(Theorem) A linear multistep method is zero stable for any ODE of the form $y' = f(x,y)$ where f satisfies the Lipschitz condition $|f(x,y) - f(x,z)| \leq L|y-z|$ Iff (if and only if) (necessary & sufficient condition) its first characteristic ⟺ polynomial has all zeros inside the closed unit disc, with any which lies on the unit circle being simple

Roots of 1st characteristic polynomial lie inside the closed unit disc & those on the circle are simple, is often called root condition.

→ multiplicity = 1

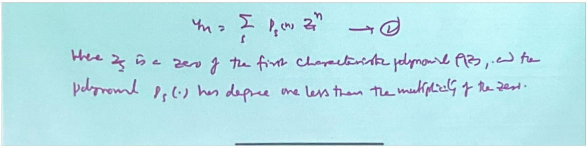


Roots of first characteristic polynomial lie in the closed unit disc and those on the circle are simple, is often called Root Condition.

Proof Necessary. Consider k-step LMM (linear multistep method), applied to $y'=0$

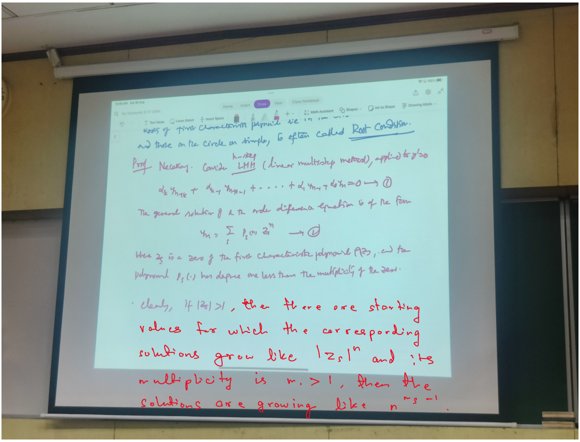
$$\alpha_k y_{n+k} + \alpha_{k-1} y_{n+k-1} + \cdots + \alpha_1 y_{n+1} + \alpha_0 y_n = 0 \longrightarrow ①$$

The general solution of k th order difference equation



$$y_n = \sum_s p_s(n) z_s^n \longrightarrow ①$$

Where z_s is a zero of the first characteristic polynomial $\rho(z)$, and the polynomial $p_s(\cdot)$ has degree one less than the multiplicity of the zero.



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$$\alpha_k y_{n+k} + \alpha_{k-1} y_{n+k-1} + \cdots + \alpha_1 y_{n+1} + \alpha_0 y_n = 0 \longrightarrow ①$$

The general solution of k th order difference equation is of the form

$$y_n = \sum_s p_s(n) z_s^n \longrightarrow ①$$

Where z_s is a zero of the first characteristic polynomial $\rho(z)$, and the polynomial $p_s(\cdot)$ has degree one less than the multiplicity of the zero.

clearly, if $|z_s|>1$, then there are starting values for which the corresponding solutions grow like $|z_s|^n$ and its multiplicity is $m_s > 1$, then the solutions are growing like $n^{m_s - 1}$.

In either case, there are solutions that grow unbounded as $n \longrightarrow \infty$, i.e., $h \longrightarrow 0$ with fixed nh.

Considering the starting data $y_0, y_1, \ldots y_{n-1}$ which give rise to such an unbiased solution y_n and the starting data $\hat{y}_0 \ldots \hat{y}_{n-1} = 0$ for which the corresponding solution is $\hat{y}_n$ with $\hat{y}_n = 0$ $\forall n$, we see that

$$|y_n - \hat{y}_n| \leq K \max\{|y_0 - \hat{y}_0|, \ldots, |y_{n-1} - \hat{y}_{n-1}|\}$$

cannot hold.

∴ If the root condition is violated, zero stability is not possible.

Sufficient - This proof is too long & technical.

LMM (linear multistep method)

Eg. $$11y_{n+3} + 27y_{n+2} - 27y_{n+1} - 11y_n = 3h\left[f_{n+3} + 5f_{n+2} + 5f_{n+1} + f_n\right]$$

is not zero stable.

First char. polynomial $\rho(z) = 11z^3 + 27z^2 - 27z - 11$ has roots $z_1 = 1$, $z_2 \approx -0.3181$, $z_3 \approx -3.1356$

$|z_3| > 1$

$$\sum_{j=0}^{k} \alpha_j y_{n+j} = h \sum_{j=0}^{k} \beta_j f(x_{n+j}, y_{n+j})$$

$$= h \sum_{j=0}^{k} \beta_j f_{n+j} \qquad \left[f_{n+j} = f(x_{n+j}, y_{n+j})\right]$$

General LMM

$$\alpha_k y_{n+k} + \alpha_{k-1} y_{n+k-1} \dots + \alpha_0 y_n$$
$$= h\{\beta_k f_{n+k} + \beta_{k-1} f_{n+k-1} + \dots + \beta_0 f_n\} \quad - ①$$

If E is the shift operator $\Rightarrow$

$$\left[\begin{array}{l} E^0 = I = y_n \\ E y_n = y_{n+1} \\ \vdots \\ E^k y = y_{n+k} \end{array}\right]$$

$$(\alpha_k E^k + \alpha_{k-1} E^{k-1} \dots \dots + \alpha_0 I) y_n$$
$$= h[\beta_k E^k + \beta_{k-1} E^{k-1} + \dots \beta_0 I] f_n$$

let

$$\rho(E) = \alpha_k E^k + \alpha_{k-1} E^{k-1} + \dots + \alpha_0 I$$
$$\sigma(E) = \beta_k E^k + \beta_{k-1} E^{k-1} + \dots + \beta_0 I$$

$$\rho(E) y_n = h\sigma(E) f_n$$
$$\rho(E) y_n - h\sigma(E) f_n = 0$$

Apply this to $y' = \lambda y$ (test eq.) $\Rightarrow$

$$\rho(E) y_n - h\sigma(E)\lambda y_n = 0$$

This is difference eq. of k^{th} order.

Its characteristic eq. is (let $\lambda h = \bar{h}$)

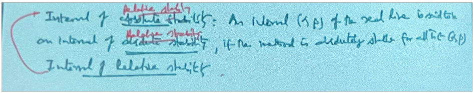
$$\rho(\xi) - \bar{h}\sigma(\xi) = 0$$

$$\pi(\xi, \bar{h}) = \rho(\xi) - \bar{h}\sigma(\xi) \qquad -(*) \Rightarrow \text{stability}$$

polynomial of ①

Let the roots of char. polynomial be $\xi_1, \xi_2, \dots, \xi_k$.

Def

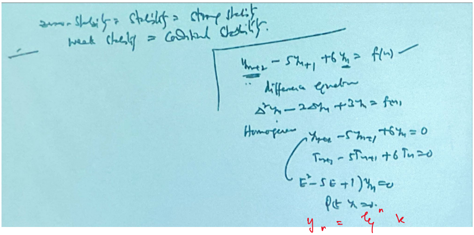
1. An LMM ① is said to be absolutely stable if $|\xi_j| < 1$ $\forall j = 1, 2, \dots k$ where ξ_j is a root of $(*)$.
2. If $|\xi_j| < |\xi_1|$ $\forall j = 2, 3 \dots k$, then ξ_1 is the root which tends to ± 1 as $h \to 0$ (relatively stable).
3. 

Interval of ~~absolute~~ Relative stability stability: An interval (α, β) of the real line is said to be an interval of ~~absolute stability~~ Relative stability, if the method is absolutely stable for all $\bar{h} \in (\alpha, \beta)$

Interval of Relative stability.

4. The method ① is said to be strongly stable if the modulus of no root of $\rho(\xi)$ [1st char. polynomial] exceeds 1 & the roots of modulus 1 must be simple.

5. Weakly stable - Method ① is said to be weakly stable if the roots with unit modulus are not simple.



zero-stability = stability = strong stability.

weak stability = Conditional stability.

$y_{n+2} - 5y_{n+1} + 6y_n = f(n)$

∴ difference equation

$\Delta^2 y_n - 2\Delta y_n + 3y_n = f(n)$

Homogeneous

$y_{n+2} - 5y_{n+1} + 6y_n = 0$

$T_{n+2} - 5T_{n+1} + 6T_n = 0$

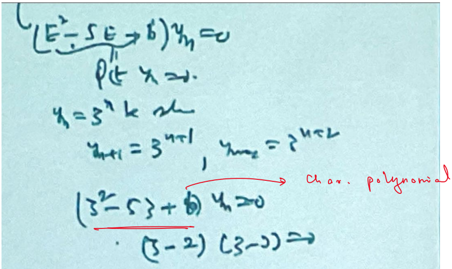
$(E^2 - 5E + 1)y_n = 0$

$\rho(E)\, y_n = 0.$

$y_n = \xi^n k$

$y_{n+1} = \xi^{n+1}$, $y_{n+2} = \xi^{n+2}$

$$(\xi^2 - 5\xi + 6)\, y_n = 0 \Rightarrow \xi = 2, 3$$



$(E^2 - 5E + 6)y_n = 0$

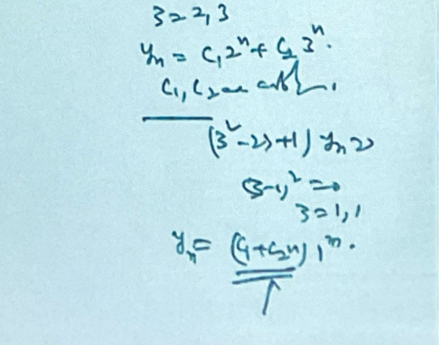
$\rho(E)\, y_n = 0.$

$y_n = 3^n k$ [illegible]

$y_{n+1} = 3^{n+1}$, $y_{n+2} = 3^{n+2}$

$(3^2 - 5\cdot 3 + 6)\, y_n = 0$ → Char. polynomial / auxiliary equation.

$(3-2)(3-3) \Rightarrow$



$z = 2, 3$

$y_n = c_1 2^n + c_2 3^n$.

c_1, c_2 are arb[illegible].

$(z^2 - 2z + 1) y_n = 0$

$(z-1)^2 \Rightarrow$

$z = 1, 1$

$y_n = \underline{(c_1 + c_2 n)}\, 1^n$.

General methods for locating intervals of absolute stability ⇒

1. Schur criterion
2. Routh-Hurwitz criterion

1. Consider the polynomial

$$\phi(r) = c_k r^k + \dots + c_1 r + c_0 \qquad c_k \neq 0, \; c_0 \neq 0$$

with complex coeff. The polynomial ϕ is said to be Schur polynomial if each of its roots r_s satisfies $|r_s| < 1$, $s = 1, \dots, k$

Let us consider the polynomial

$$\hat{\phi}(r) = \bar{c}_0 r^k + \bar{c}_1 r^{k-1} + \dots + \bar{c}_k$$

where $\bar{c}$ denotes complex conjugate of c.

Further, we denote

$$\phi_1(r) = \frac{1}{r}\left[\hat{\phi}(0)\,\phi(r) - \phi(0)\,\hat{\phi}(r)\right]$$

Degree of $\phi_1 \leq k-1$

Theorem (Schur's criterion) The polynomial ϕ is Schur polynomial iff $|\hat{\phi}(0)| > |\phi(0)|$ and ϕ_1 is a Schur polynomial.

2. Consider the mapping $z = \frac{r-1}{r+1}$ of the open disc $|r| < 1$ of complex r-plane to left half plane $Re(z) < 0$ of complex z-plane.

Inverse map ⇒ $r = \frac{z+1}{1-z}$

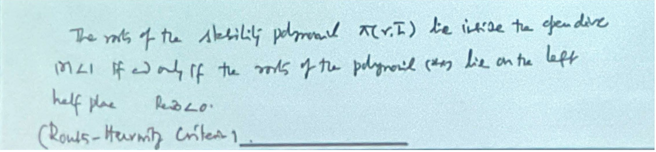
Under this transformation, the function

$\pi(r, \bar{h}) = \rho(r) - \bar{h}\,\sigma(r)$ becomes

$$\pi(z, \bar{h}) = \rho\left(\frac{1+z}{1-z}\right) - \bar{h}\,\sigma\left(\frac{1+z}{1-z}\right) \qquad (\bar{h} = \lambda h)$$

Multiply by $(1-z)^k \Rightarrow$ We get a polynomial of the form

$$a_0 z^k + a_1 z^{k-1} + \dots + a_k \qquad - (**)$$



The roots of the stability polynomial $\pi(r,\bar{h})$ lie inside the open disc $|r|<1$ iff and only iff the roots of the polynomial (**) lie on the left half plane $\operatorname{Re} z < 0$.

(Rouths-Hurwitz Criteria)

The root of (**) lie in the open left half plane iff all the principal minors of the $k \times k$ matrix

$$Q = \begin{bmatrix} a_1 & a_3 & a_5 & \cdots & a_{2k-1} \\ a_0 & a_2 & a_4 & \cdots & a_{2k-2} \\ 0 & a_1 & a_3 & \cdots & a_{2k-3} \\ 0 & a_0 & a_2 & \cdots & a_{2k-4} \\ \vdots & & & & \\ 0 & 0 & \cdots & \cdots & a_k \end{bmatrix}$$

determinants of the top left square matrices are the principal minors.

are +ve and $a_0 > 0$, we assume $a_j = 0$ if $j > k$.

1) For $k = 2$ $\quad a_0 > 0,\ a_1 > 0,\ a_2 > 0$

2) For $k = 3$ $\quad a_0 > 0,\ a_1 > 0,\ a_1 a_2 - a_3 a_0 > 0$

3) For $k = 4$ $\quad a_0 > 0,\ a_1 > 0,\ a_1 a_2 - a_3 a_0 > 0,$

$\qquad a_1 a_2 a_3 - a_0 a_3^2 - a_4 a_1^2 > 0$

This represents necessary & sufficient condition for ensuring that the roots of (**) lie on the open left half plane.

[Asymptotic stability → A-stability]

$$\sum_{j=0}^{k} \alpha_j y_{n+j} = h \sum_{j=0}^{k} \beta_j f_{n+j} \qquad - \text{(I)}$$

$$\Rightarrow \alpha_k\, y(x+kh) + \alpha_{k-1}\, y(x+(k-1)h) + \dots + \alpha_0\, y(x)$$

$$- h\left[\beta_k\, y'(x+kh) + \beta_{k-1}\, y(x+(k-1)h) + \dots + \beta_0\, y'(x)\right]$$

$$= \underline{L}\,(y(x), h) \qquad - ①$$

↳ This may be regarded as a linear operator that acts on any diff. function $y(x)$.

Expand $L(y(x), h)$ in powers of h ⇒

$$y(x+mh) = y(x) + mh\, y'(x) + \frac{(mh)^2}{2!}\, y''(x) + \dots$$

$$h y(x+mh) = hy(x) + mh^2\, y'(x) + \frac{m^2h^3}{2!}\, y''(x) + \dots$$

It turns out that

$$L(y(x), h) = c_0\, y(x) + c_1 h\, y'(x) + \dots + c_v h^v y^{(v)}(x) \dots \qquad - ②$$

Here the coeff. $c_0, c_1 \dots$ do not depend on the choice of function $y(x)$.

$$c_0 = \alpha_0 + \alpha_1 + \dots + \alpha_k$$

$$c_1 = \alpha_1 + 2\alpha_2 + \dots + k\alpha_k - (\beta_0 + \beta_1 + \dots + \beta_k)$$

$$\vdots$$

$$c_q = \frac{1}{q!}\left(\alpha_1 + 2^q\alpha_2 + \dots + k^q\alpha_k\right) - \frac{1}{(q-1)!}\left(\beta_1 + 2^{q-1}\beta_1 + \dots + k^{q-1}\beta_k\right)$$

Order A given diff. operator ② is said to be order p if $c_0 = c_1 = \dots = c_p = 0$ but $c_{p+1} \neq 0$

① is equivalent to

$$\alpha_k\, y_{n+t+k} + \dots + \alpha_0\, y_{n+t} = h\left(\beta_k f_{n+t+k} + \dots + \beta_0 f_{n+t}\right) \qquad - ③$$

Here t is any integer. We may associate ③ with the diff. operator.

$$L_t\left[y(x), h\right] = \alpha_k y_{n+t+k} + \dots + \alpha_0 y_{n+t} - h\left[\beta_k f_{n+t+k} + \dots + \beta_0 f_{n+t}\right]$$

$$L_t\left[y(x), h\right] = \alpha_k y(x+(t+k)h) + \dots + \alpha_0 y(x+th) - h\left[\beta_k f(x+(t+k)h) + \dots + \beta_0 f(x+th)\right] \quad - ④$$

and we define the order as the order of the 1st non vanishing term with Taylor's expansion in powers of h minus 1.

It is an imp. fact that order p & constant C_{p+1} do not depend on t. For, expanding ④ in powers of h is equivalent to expanding ② in h, just, $y(x)$ is now replaced by $y(x+th)$.

We find that if L is of order p, then $L_t(y(x), h) = L(y(x+th), h) = C_{p+1} h^{p+1} y^{p+1}(y+th) + O(h^{p+2})$

since $y(x)$ is assumed sufficiently diff.,
$$y^{(p+1)}(x+th) = y^{(p+1)}(x) + O(h)$$

$$L_t(y(x+th), h) = C_{p+1} h^{p+1} y^{(p+1)}(x) + O(h^{p+2})$$

$\therefore$ Order remains the same.

Error constant of ① is $c = \dfrac{C_{p+1}}{\beta_0 + \beta_1 + \dots \beta_k}$

Consistency If $p \geq 1$, method ① is said to be consistent.

i.e., $C_0 = C_1 = 0$, or $\rho(1) = 0$
and $\rho'(1) = \sigma(1)$

[ρ and σ are 1st & 2nd char. poly.]

$p > 1 \Rightarrow$ Condition of consistency.

Truncation error of the LMM ① is defined as $\Rightarrow$

$$T_n = \sum_{j=0}^{k} \alpha_j y(x_{n+j}) - \sum_{j=0}^{k} \beta_j y'(x_{n+j}) \quad - (II)$$

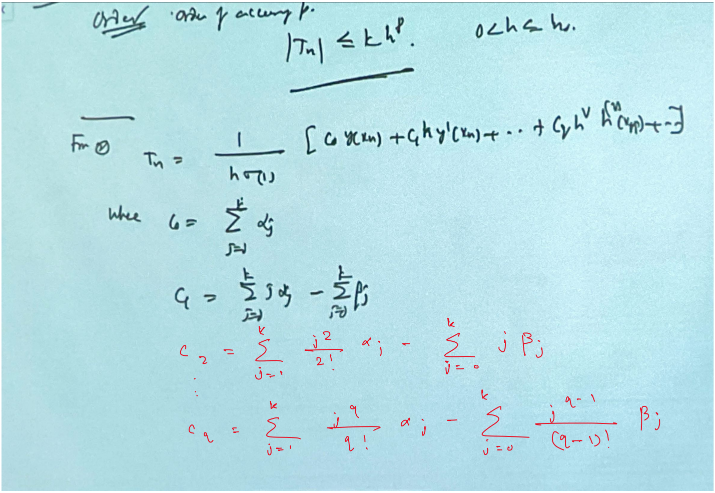
$$h \sum_{j=0}^{k} \beta_j$$

$$\sigma(1) = \sum \beta_j \neq 0$$

The numerical scheme ① is said to be consistent with the diff. equation $y' = f(x, y)$ if the TE ② is s.t. for any $\varepsilon > 0$, $\exists$ $h(\varepsilon)$ for which

$$|T_n| < \varepsilon \quad \text{for} \quad 0 < h < h(\varepsilon)$$

and any $k+1$ points $(x_n, y_n) \ldots (x_{n+k+1}, y_{n+k+1})$ or any solution of IVP $y' = f(x, y)$, $y(x_0) = y_0$



Order: order of accuracy p.

$$|T_n| \le k h^p. \qquad 0 < h \le h_0.$$

From ① $T_n = \dfrac{1}{h\sigma(1)}\left[C_0 y(x_n) + C_1 h y'(x_n) + \cdots + C_q h^q y^{(q)}(x_n) + \cdots\right]$

where $C_0 = \sum_{j=1}^{k} \alpha_j$

$$C_1 = \sum_{j=1}^{k} j\alpha_j - \sum_{j=0}^{k} \beta_j$$

$$C_2 = \sum_{j=1}^{k} \frac{j^2}{2!} \alpha_j - \sum_{j=0}^{k} j \beta_j$$

$\vdots$

$$C_q = \sum_{j=1}^{k} \frac{j^q}{q!} \alpha_j - \sum_{j=0}^{k} \frac{j^{q-1}}{(q-1)!} \beta_j$$